

SECTION 01009-C

F16 EHSS - 管線標示作業標準

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1. 目的 Purpose:

建立各類管線配置時標示作業之明確規定，以使管線系統易於辨識管理，提高工廠內人員及設備之安全。

Establish clear provisions for labeling various pipelines so that the piping system is easy to identify and manage, and improve the safety of personnel and equipment in the factory.

2. 適用範圍 Scope:

適用於公司化學品、氣體、水、排氣、排水管線、消防管線及電力系統等。

Suitable for the company's chemical, gas, water, exhaust, drainage, and firefighting pipelines, and electrical systems.

3. 名詞定義 Definition:

3-1 標示顏色：依管線內輸送物質之危害性或特性分類之代表顏色。

Colored labeling: color label the pipes according to the hazard level or characteristic type of the materials the pipe is carrying.

3-2 標示內容：管內輸送物質說明，包括物質名稱及流向。

Labeling content: describe the material the pipe is transporting, including material name and flow direction.

3-3 標示尺寸：依管徑大小規範標示尺寸，以清楚辨識標示內容。

Labeling Size: label the size according to the pipes' dimensions, and clearly identify label content.

3-4 標示之位置及間距：依管線及設備於作業環境之設置，張貼標示之位置及間距。

Location and spacing of the labels: the label position and spacing shall be according to the pipeline and installation placement in the work environment.

3-5

4. 參考文件 Reference:

4-1 危害性化學品標示及通識規則

Regulation of Labeling and Hazard Communication of Hazard Materials

4-2 特定化學物質危害預防標準

Specific chemical prevention standard

4-3 職業安全衛生設施規則

Occupational safety and health facility regulation

4-4 中國國家標準 CNS9329 Z1025

Chinese National Standard CNS9329 Z1025

4-5 J4TQ6QXK32ES-850338125-4358 危害通識執行標準

Implementation Standard of Hazard Communication on Hazardous Materials

4-6 09005aef8662e543 Corporate EHS TMAH Safety Standard

4-7 09005aef84daeac6 TMAH Facility of Record Standard

5. 權責 Responsibility:

5-1 全廠各部門及到廠施工廠商均須遵守。

All the departments in the plant and construction vendors who come on site are required to comply.

6. 作業內容 Content:

6-1 標示顏色

Label Colors

6-1-1 標示之顏色係依管線內容物危害性或特性分類，利用標示底色及文字顏色搭配予以區分，分類原則參考附件 8-1-1。

The colors of the labels are according to the hazard level and characteristic type of the content carried by the pipes. The labels should use background and text colors for distinguishing. For classification principle, reference Appendix 8-1-1.

6-1-1-1 特殊氣體、一般氣體、化學品、排氣、排水此類管線標示之底色參考危害性化學品標示及通識規則中危害性分類之規定，分為易燃性、腐蝕性或含 1%以上四甲基氫氧化銨、毒性、非危害性製程原物料、氧化性、禁水性及其他。

For special gases, regular gas, chemicals, exhaust, and drainage pipe labeling background color, reference the regulations in the Regulation of Labeling and Hazard Communication of Hazard Materials, which is divided into flammable, corrosive or contain above 1% TMAH, toxic, non-HPM, oxidizing, react with water, and other.

6-1-1-2 水（補給水、純水及冷卻水）

water (water supply, pure water, and cooling water)

此類管線標示之底色為藍色，文字為黑色。

Such pipeline labels use a blue background with black text.

6-1-1-3 熱水

Hot water

此類管線標示之底色為紅色，文字為白色。

Such pipeline labels use a red background and white text.

6-1-1-4 消防管線

Firefighting pipelines

此類管線均將管線漆以紅色油漆，標示底色為紅色，文字為白色。

Such pipelines are painted red, with labels using a red background with white text.

6-1-1-5 電力系統

Power system

依其電壓值，< 600V 者為低電壓，> 600V 者為高電壓，於系統盤面/Cable Tray/Bus Way 等電力供應/傳輸區域明顯易見處，張貼警告標示。

Based on their voltage. Less than 600 V is low voltage, over 600 V is high voltage. Warning labels shall be placed at an easily visible location at the electrical supply system panel/cable trays/bus way, or other power transmission/supply area.

6-2 標示內容

Label Content

6-2-1 名稱之識別

Name Identification

6-2-1-1 記號：所有管線應標註管內輸送物質中英文名稱（如：磷化氫 Phosphine、三氟化硼 Boron trifluoride、氯 Chloride）。

Labels: all pipelines should identify the material Chinese and English name it is transporting (example: Phosphine, Boron trifluoride, Chloride).

6-2-1-2 表示方法：標示名稱之文字顏色應符合附件 8-1-1 規定。

Labeling method: The Chinese text in the label name should meet the requirement of appendix 8-1-1.

6-2-2 流向之識別

Flows direction identification

6-2-2-1 記號：流向之識別應使用箭頭。

Label: Arrows should be used in the identification of the flows direction.

6-2-2-2 表示方法：箭頭之識別顏色應與文字相同。

Indication method: The color of arrow indication should be the name with text.

6-2-3 含 1%以上四甲基氫氧化銨廢液識別

Contain above 1% TMAH waste water identification

6-2-3-1 含有 1%以上四甲基氫氧化銨之廢液，除標註 6-2-1 及 6-2-2 之名稱及流向識別外，應增加附件 8-1-3 之警告標示，以明確標示

內含四甲基氫氧化銨。

Except 6-2-1 and 6-2-2, name and flow direction identification, the waste water contained above 1% TMAH shall add the warning label, appendix 8-1-3, to clear identify it contain TMAH.

6-3 標示尺寸

Label Dimensions

6-3-1 標示尺寸以能明顯識別為原則，可依管徑尺寸調整標示，但不得小於下述規範之最小尺寸。

The size of label shall be clear identified and could be modify according to piping size, but shall not be smaller than the required size below.

6-3-2 依管徑大小分為三種，下述標示尺寸為**最小尺寸**，若字數較多，長度可作適當調整。

There are three types of pipes according to its size. The labeling size described below is of the **smallest** size. If there are more words, the length can be suitability adjusted.

6-3-2-1 管徑 25A 以下（含 25A，約 1 吋）：標示尺寸如附件 8-1-2 圖示大小（**130 mm×20 mm**），英文字之字型為 Arial. 粗體，中文字之字型為**微軟正黑體**. 粗體。

Pipe diameter below 25A (including 25A, **about 1 inch**): label size should be as the picture in Appendix 8-1-2 icon (**130 mm x 20 mm**). The English font should be Arial in bold. The Chinese font should be Microsoft **JhengHei** with bold script.

6-3-2-2 管徑**超過 25A**（約 1 吋）、**小於 75A**（約 3 吋）：標示尺寸如附件 8-1-2 圖示大小（**155mm×50 mm**），英文字型為 Arial. 粗體，中文字型為**微軟正黑體**.粗體，若字數較多，可中英文分列為兩行。

Diameter **above 25A (about 1 inch), less than 75A (about 3 inch)**: label size should be as the picture in Appendix 8-1-2. The label size should be **155mm × 50mm**. The English font should be Arial, in bold. The Chinese font should be in **Microsoft JhengHei** and in bold. If there are more words, the English text can be divided into two rows with the size in 32 pt. bold.

6-3-2-3 管徑 **75A** 以上（含 75A，約 3 吋）：標示尺寸如附件 8-1-2 圖示大小（**250 mm×100 mm**），英文字型為 Arial.粗體，中文字型為**微軟正黑體**.粗體。

For pipe diameters above **75A (including 75A, about 3 inch)**: the label should be as the picture in Appendix 8-1-2. The text and arrow label size should be **250 mm × 100 mm**. The English should be in Arial. bold, and the Chinese font should be **Microsoft JhengHei** . bold.

- 6-3-3 電力系統標示尺寸：如附件 8-1-2 圖示，文字之標示尺寸為 300mm × 80mm，英文字之字型為 Arial 78 pt. 粗體，中文字之字型為**微軟正黑體** 120 pt. 粗體。

The power system labeling size: as show in the picture of the appendix. The text label size should be 300mm × 80mm. The English font should be in Arial 78 pt. bold, and the Chinese font should be in MingLiU 120 pt. bold.

6-4 標示之位置及間距

Label Position and Spacing

- 6-4-1 管線標示應張貼於作業人員可明顯易見處且**不可被遮蔽，如通道處、檢修處**。

Pipeline labels of all types should be placed where it can easily be seen by operating personnel, and could not be covered, e.g. walkway, inspection point.

- 6-4-2 標示內容之名稱及流向應相鄰張貼，不可分開。

Label content's name and flow direction should be adjacent to each other and not be separated.

- 6-4-3 標示之間距於無閥件及通過牆之直管至多為五公尺。

The spacing of the labels should be no more than 5 meters from the no-valve or the wall penetration straight pipe.

- 6-4-4 若管線為彎管及在閥件、牆之兩側**一公尺內**均應作標示；於一公尺內有兩處以上彎管或閥件時，若可清楚辨識且不造成混淆者，得僅標示一處予以代表。

If the pipe bends, then there should be a label at the valve and both sides **within 1 meter** of the wall. If there are more than two bends or valves within one meter, only one location needs to be labeled as a representative as long as it can be identified clearly and not cause confusion.

- 6-5 若有此規則未規範之化學物質，工程承辦單位於標示製作前，應與 EHSS 討論以確認應標示之規格。

If there are chemicals that have not been regulated by these rules, then the engineering department undertaking the labeling should consult EHSS regarding the correct label specifications prior to labeling.

7. 作業流程圖 Flowchart:

無 N/A

8. 附件表單 Appendix:

8-1 附件 Appendix

- 8-1-1 管線標示顏色分類。

Pipeline colored label classification.

8-1-2 管線標示規格圖示。

Pipeline labeling specification pictures.

8-1-3 含 1%以上四甲基氫氧化銨之廢液警告標示

Warning label of waste water contained above 1% TMAH

8-2 表單 Form

無 N/A

附件 8-1-1 管線標示顏色分類

管線內容物		標示底色/文字顏色
特殊氣體、一般氣體、 化學品、排氣、排水	易燃性	紅/黑
	腐蝕性或含 1%以上四甲基氫氧化銨	橙/黑
	毒性	黃/黑
	非危害性製程原物料	黑/白
	氧化性	紫/黑
	禁水性	藍/白
	其他(others)	白/黑
水 (補給水、純水、冷卻水)		藍/黑
熱水(hot water)		紅/白
消防管線系統		紅/白
電力系統	低電壓 $\leq 600V$	黃/黑
	高電壓 $> 600V$	黃/紅

Appendix 8-1-1 Pipeline colored label classification

Pipeline content		Label background color / text color
special gas, general gas, chemicals, exhaust, drainage	Flammable	red/black
	Corrosive, or contain above 1% TMAH	orange/black
	Toxic	yellow/black
	Non-HPM	black/white
	Oxidative	purple/black
	React with water	blue/black
	Others	white/black
Water (supply water, pure water and cooling water)		blue/black
Hot water		red/white
Firefighting pipeline system		red/white
Power system	low voltage $\leq 600V$	yellow/black
	high voltage $> 600V$	yellow/red

附件 8-1-2 管線標示規格圖示

管線標示規格表：

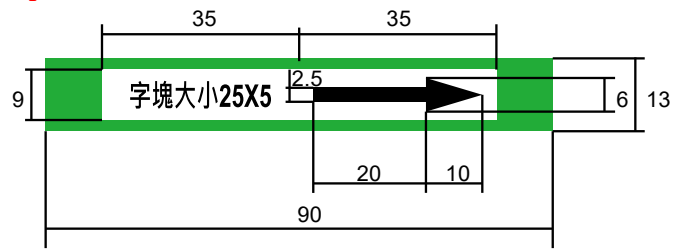
管徑	標示尺寸	英文字型	中文字型
特殊管路(3/8、2/8 英吋)	90 × 13 mm	Arial, 粗體	微軟正黑,粗體
25A 以下 (含 25A , 約 1 吋)	130 × 20 mm	Arial, 粗體	微軟正黑體, 粗體
超過 25A (約 1 吋)、 小於 75A (約 3 吋)	155 × 50 mm	Arial, 粗體	微軟正黑體, 粗體
75A 以上 (含 75A , 約 3 吋)	250 × 100 mm	Arial, 粗體	微軟正黑體, 粗體
電力系統	300 × 80 mm	Arial, 粗體	微軟正黑體, 粗體

Pipelining labeling specification list:

Pipe size	Label size	English text	Chinese text
Special size (3/8 、 2/8 inch)	130 × 20 mm	Arial, bold	Microsoft JhengHei, , bold
Below 25A (including 25A , about 1 inch)	130 × 20 mm	Arial, bold	Microsoft JhengHei, , bold
Over 25A (about 1inch), below 75A (about 3 inch)	155 × 50 mm	Arial , bold	Microsoft JhengHei, , bold
Over 75A (including 75A, about 3inch)	250 × 100 mm	Arial, bold	Microsoft JhengHei, , bold
Electrical system	300 × 80 mm	Arial, bold	Microsoft JhengHei, bold

特殊小管路(3/8、2/8 英寸)

Special size (3/8、2/8 inch)



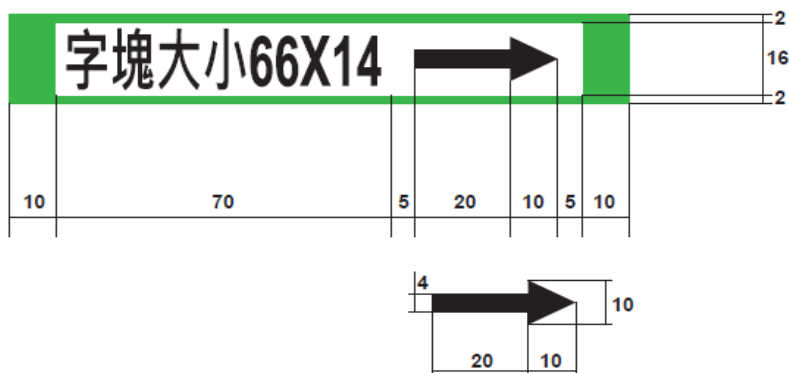
範例:

Example:



管徑 25A 以下 (含 25A, 約 1 吋) :

Pipe Size: 25A and below (including 25A, about 1 in):



範例:

Example:



管徑超過 25A (約 1 吋)、小於 75A (約 3 吋)：

Pipe Size: 25A (about 1 in) ~ 75A (about 3 in)



範例

Example:



管徑75A以上 (含75A，約3吋)：

Pipe Size: 75A and above (including 75A, about 3 in):



範例

Example:



附件 8-1-2 電力系統標示：

Electrical system labeling:

標示尺寸:300 × 80 mm Arial, 78 pt, 粗體,微軟正黑體, 120 pt,粗體.

The label size should be :300 × 80 mm Arial, 78 pt, bold Microsoft JhengHei, 120 pt, bold



電管(EMT)標示：

Electric tube (EMT) labeling:

外框100×30 mm,內框,內框90×20 mm, 文字方塊85×15 mm, 英文字型為Arial.粗體，中文字型為微軟正黑體.粗體。

External frame 100 × 30 mm, inner frame 90×20 mm, text box 85×15 mm .The English should be in Arial. bold, and the Chinese font should be Microsoft JhengHei . bold.



附件 8-1-3 含 1%以上四甲基氫氧化銨之廢液警告標示

Warning label of waste water contained above 1% TMAH

